Biographical Irreducibility in Quantum Mechanics: Path Inequivalence Beyond State Equivalence

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I demonstrate that quantum systems implementing the same unitary operation and sharing identical final states can nonetheless exhibit measurably distinct behavior under environmental decoherence. This arises when the underlying dynamical trajectories—the "biographies" of the processes—differ geometrically in state space even though the endpoints coincide. I construct explicit "quantum twins" that satisfy:

1. Same unitary: $U_A = U_B$
2. Same final state: $\rho_A = \rho_B$
3. Distinct trajectories: $K_A \neq K_B$
4. Decoherence-dependent divergence: $\rho'_A \neq \rho'_B$

Through mathematical analysis, Bloch-sphere geometry, time-normalized evolution, multiple physical noise models, multi-qubit extensions, bootstrap statistics (observed Frobenius-norm difference ≈ 0.244, 95(Here, 0.244 refers specifically to the Frobenius norm difference $\|\rho'_A - \rho'_B\|_F$. Other metrics such as purity difference ΔP are introduced and defined later.), and robustness tests against initial correlations (50 random pure states, 50 mixed states, 6 basis states), I show that biographical information is physically relevant and irreducible to instantaneous state descriptions. I provide an experimental protocol for photonic implementation and discuss implications for the ontology of quantum processes.

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1 Introduction

1.1 Motivation

In conventional quantum mechanics, the state $\rho(t)$ is taken to encode all physically accessible information at a given time. This implicitly assumes that two systems with identical states are physically indistinguishable, even if they reach those states through different histories.

However, this assumption has never been rigorously validated for open quantum systems. Decoherence couples a system to an environment, and this coupling depends on the system's *trajectory*, not merely its instantaneous state. This raises a fundamental question about the completeness of state-based descriptions: if two processes yield identical final states but traverse different paths through Hilbert space, does this *biographical* difference carry physical content? Recent work on non-Markovian and process-tensor descriptions has highlighted precisely these concerns [1, 2].

The distinction is not merely semantic. In classical mechanics, two trajectories reaching the same endpoint are operationally equivalent—only the final state matters for prediction. But quantum mechanics, with its intricate relationship between system and environment through entanglement, may tell a different story. The path itself—the continuous unfolding of the quantum process—may leave an indelible signature detectable through decoherence.

The central question I address is:

> **Can two quantum processes that share the same initial state, same final state, and same logical operation still be physically distinguishable due solely to differences in their dynamical histories?**

I show that the answer is **yes**, and that this biographical irreducibility has measurable empirical consequences.

1.2 Main Result

I construct quantum processes ("twins") such that:

$$U_A = U_B, \quad \rho_A = \rho_B, \quad K_A \neq K_B, \tag{1}$$

yet after decoherence,

$$\rho'_A \neq \rho'_B. \tag{2}$$

This proves that the "biography" K—the entire dynamical path through state space—contains physically relevant information not captured by ρ.

1.3 Implications

This challenges the assumption of state completeness and suggests that process-based ontologies (e.g., process tensors, quantum combs) reflect deeper physical structure than commonly appreciated.

2 Theoretical Framework

2.1 Biographical Information

The **biography** of a quantum system is the full trajectory:

$$K = \{\rho(t) : t \in [0, T]\}. \tag{3}$$

For unitary evolution:

$$\rho(t) = U(t)\rho_0 U^\dagger(t), \quad U(t) = \mathcal{T} \exp\left(-i \int_0^t H(t')dt'\right). \tag{4}$$

I use the geometric witness:

$$W_{\text{bio}} = \int_0^T \|\dot{\rho}(t)\| dt. \tag{5}$$

2.2 Path Inequivalence Under SU(2)

The same rotation $R_y(\theta)$ can be achieved via:

$$R_y(\theta) = e^{-i\theta\sigma_y/2} \tag{6}$$

or through the decomposition:

$$R_z(\phi)R_x(\theta)R_z(-\phi), \quad \phi = \frac{\pi}{2}. \tag{7}$$

These operations share the same final state, but follow different Bloch-sphere curves, yielding different W_{bio}. For open-system modeling and the master-equation approaches used to simulate decoherence we follow standard treatments [3]. The decomposition and gate identities used here are standard in quantum computation and are discussed in textbooks and circuit-architecture literature [4, 5].

3 Construction of Quantum Twins

Initial state:

$$|\psi_0\rangle = |0\rangle. \tag{8}$$

Twin A (Direct Path): Apply $R_y(\theta)$ in a single rotation.
Twin B (Decomposed Path): Apply $R_z(\frac{\pi}{2})R_x(\theta)R_z(-\frac{\pi}{2})$ in three sequential rotations.
Verification

$$\|U_A - U_B\| = 1.96 \times 10^{-16}, \quad \|\rho_A - \rho_B\| = 2.57 \times 10^{-16}. \tag{9}$$

Path Lengths

$$W_A = 1.047, \quad W_B = 1.331, \quad \frac{W_B - W_A}{W_A} = 27.1\%. \tag{10}$$

Result: Identical endpoints, different geometry.

4 Computational Demonstrations

All results below are fully reproducible via `qmsg_complete.py`.

4.1 v2.0 — Mathematical Proof

Using 20-step Bloch trajectories, I verified:

- Operations identical to machine precision ($\|U_A - U_B\| < 10^{-15}$)
- Final states identical ($\|\rho_A - \rho_B\| < 10^{-15}$)
- Paths inequivalent (27.1% difference in length, see Fig. S1)

This establishes the **geometric irreducibility** of K to ρ.

4.2 v3.1 — Time Normalization

Critical test: Does the effect arise from execution time differences?

I enforced $T_A = T_B = 1.0$ and tested two decoherence models:
Conclusion: Effect persists with time control. Difference is **not** a timing artifact. Purity difference (path model): $\Delta\text{Purity} = 0.145$.

Model	γ_A	γ_B	$\|\rho'_A - \rho'_B\|$
$\gamma \propto T$	0.150	0.150	1.54×10^{-16}
$\gamma \propto W$	0.157	0.356	**0.244**

Table 1: Time-normalization test results (Stress Test configuration with amplified coupling to demonstrate qualitative effect independence from time).

4.3 v3.2 — Physical Noise Models

I tested four physically motivated noise environments:

Noise Model	$\|\rho'_A - \rho'_B\|$	Δ Purity
White	0.244	0.145
Ohmic	0.293	0.138
Super-Ohmic	0.195	0.141
1/f	0.393	0.219

Table 2: Robustness across physical noise models.

Conclusion: Effect is **robust across all physical noise models**.

4.4 v3.3 — Multi-Qubit Scaling

Extension to 2-qubit systems:

$$U_A = R_y^{(1)}(\theta) \otimes I, \quad U_B = [R_z R_x R_z]^{(1)} \otimes I \tag{11}$$

Results:

- 1-qubit divergence: 0.244
- 2-qubit divergence: 0.200

Conclusion: Effect **scales naturally** with Hilbert space dimension.

4.5 v3.4 — Statistical Significance

Bootstrap resampling (n = 1000) with measurement noise:

- **Observed difference:** 0.244
- **Bootstrap mean:** 0.245
- **Bootstrap std:** 0.016
- **95% CI:** [0.217, 0.275]
- **p-value:** $< 10^{-6}$
- **z-score:** 21.65σ

Note: Here, the quoted $z \approx 21.65$ refers to the bootstrap signal-to-noise ratio rather than to a null-hypothesis significance test. It measures robustness of the observed divergence under resampling, not discovery-level statistical significance. Accordingly, bootstrap z-scores should not be compared directly to 5σ discovery thresholds.

Conclusion: The effect is extremely robust under bootstrap resampling, though this should not be interpreted as a $> 5\sigma$ null-hypothesis discovery claim.

4.6 v3.5 — Initial Correlations Robustness

Critical concern: Could initial system-environment correlations explain the effect?

I tested 106 distinct initial conditions:

- **50 random pure states** (Haar distribution)
- **50 random mixed states** (controlled purity)
- **6 computational basis states** ($|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle, |-i\rangle$)

Pure States Results:

- Mean divergence: 0.223 ± 0.054
- Min: 0.104, Max: 0.282

Mixed States Results:

- Initial purity range: 0.786 ± 0.101
- Mean divergence: 0.171 ± 0.059

Basis States Results:

State	Divergence $\|\rho'_A - \rho'_B\|$
$\|0\rangle$	0.244
$\|1\rangle$	0.244
$\|+\rangle$	0.141
$\|-\rangle$	0.141
$\|+i\rangle$	0.282
$\|-i\rangle$	0.282

Table 3: Robustness across computational basis states.

Conclusion: Effect is detectable across the majority of initializations, with some dependence on initial state structure that merits further investigation.

5 Experimental Proposal

5.1 Photonic Implementation

Platform: Polarization-encoded single photons

Protocol:

1. **Prepare** $|H\rangle$ (horizontal polarization)
2. **Implement Twin A:** Single wave plate at angle $\theta/2$
3. **Implement Twin B:** Three wave plates (decomposition)
4. **Verify equality:** Quantum state tomography confirms $\rho_A = \rho_B$
5. **Introduce noise:** Controlled birefringent medium
6. **Measure divergence:** Tomography + purity estimation

Prediction: Twin B exhibits greater purity loss proportional to path difference.

5.2 Resource Requirements

- Single-photon source (SPDC or heralded)
- Wave plates: $\lambda/2$ and $\lambda/4$ (precision $< 1°$)
- Birefringent noise source (controlled glass)
- Detection: APDs + coincidence counting
- Statistics: $\sim 10^4$ counts per configuration

Feasibility: Standard quantum optics laboratory. A detailed implementation protocol, including error budget analysis, timeline, and resource requirements, is provided in Appendix E.

6 Discussion

6.1 Foundational Implications

6.1.1 State Incompleteness

The density matrix $\rho(t)$ does not contain all physically relevant information. In open-system evolution, the system's history K has measurable consequences not captured by instantaneous states.

6.1.2 Process Ontology

These results are consistent with interpretations in which dynamical evolution—not only instantaneous states—may carry physically relevant structure. This perspective is compatible with certain process-oriented views in the foundations of quantum mechanics.

6.1.3 Process Tensor Connection

The biographical witness W_{bio} is naturally related to process tensor formalism. Future work will derive rigorous bounds connecting W to quantum Choi-Majer information (QCMI).

6.2 Limitations

Current work:

- Computational demonstrations (not yet experimental)
- Limited to phase damping noise models tested
- 1–2 qubit systems only
- Sensitivity to initialization observed in v3.5 requires further investigation
- No formal impossibility theorem yet

These limitations do not undermine the core result but indicate directions for extension.

6.3 Future Directions

1. **Experimental validation** (photonics, trapped ions)
2. **Derive formal impossibility theorem** for K $\to \rho$ compression
3. **Extend to general CPTP maps** beyond unitary twins
4. **Higher-dimensional systems** (3+ qubits, continuous variables)
5. **Investigate initialization dependence** observed in v3.5
6. **Applications:**
   - Biography-aware error correction
   - Quantum metrology exploiting K
   - Thermodynamic implications
   - Process-based quantum algorithms

7 Conclusion

I have demonstrated that **quantum biographical information K is operationally relevant and irreducible to state information** ρ.

Through comprehensive validation—mathematical proof, time normalization, physical noise models, multi-qubit scaling, statistical significance ($p < 10^{-6}$, $z = 21.65\sigma$), and robustness across 106 initial conditions—I establish that:

> **Two processes implementing the same unitary with the same final state can diverge measurably under decoherence if their trajectories differ.**

This proves that the "middle" of quantum evolution carries physical content absent from the "extremes" (initial and final states).

Testable prediction: Different gate decompositions yield different decoherence signatures in real quantum hardware.

Philosophical implication: Quantum mechanics requires process-centric descriptions for completeness.

The trajectory matters fundamentally—not just for calculation, but for physical reality.

Acknowledgments

I thank the researchers whose foundational work in open quantum systems and process tensor theory provided the conceptual framework for this investigation.

References

[1] Felix A Pollock, César A Rodríguez-Rosario, Thomas Frauenheim, Mauro Paternostro, and Kavan Modi. Non-markovian quantum processes: Complete framework and efficient characterization. *Physical Review A*, 97(1):012127, 2018.

[2] Simon Milz, Kavan Modi, Parveen Sakuldee, Matthias Krumm, and Časlav Brukner. Quantum stochastic processes and quantum non-markovian phenomena. *PRX Quantum*, 2(3):030201, 2021.

[3] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, 2002.

[4] Michael A Nielsen and Isaac L Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.

[5] Giulio Chiribella, Giacomo Mauro D'Ariano, and Paolo Perinotti. Quantum circuit architecture. *Physical Review Letters*, 101(6):060401, 2008.

A Complete Reproducible Code

All results are generated by `qmsg_complete.py` (600 lines, fully documented):
Contents:

- v2.0: Mathematical proof ($K \neq \rho$ geometry)
- v3.1: Time-normalization test
- v3.2: Physical noise models (White, Ohmic, Super-Ohmic, 1/f)
- v3.3: Multi-qubit scaling (2-qubit extension)
- v3.4: Bootstrap statistics (n=1000)
- v3.5: Initial correlations robustness (106 tests)
- Master figure generation (6-panel publication figure)

Runtime: $\sim$30 seconds on standard hardware.
Availability: https://github.com/AndersonCRodrigues/qmsg-biographical-irreducibility

B Figure Captions

Figure 1: Complete Demonstration of Biographical Irreducibility

(A) Bloch Sphere Trajectories
Blue solid line: Twin A (direct rotation). Red dashed line: Twin B (three-step decomposition). Green star: initial state $|0\rangle$. Purple square: final state (identical for both). Both paths reach the same endpoint but follow different geodesics.

(B) Biographical Witness
Path length comparison: $W_A = 1.047$, $W_B = 1.331$ (27.1% difference). Despite operational equivalence, geometric structure differs measurably.

(C) Time-Normalization Test
Left bar (green): Model $\gamma \propto T$ shows no divergence ($\sim 10^{-16}$). Right bar (red): Model $\gamma \propto W$ shows large divergence (0.244). Effect persists under strict time control, proving it's not a timing artifact.

(D) Physical Noise Models
Divergence across four physical environments: White (0.244), Ohmic (0.293), Super-Ohmic (0.195), 1/f (0.393). Effect is robust regardless of spectral density.

(E) Multi-Qubit Scaling
Blue bar: 1-qubit divergence (0.244). Purple bar: 2-qubit divergence (0.200). Effect scales naturally with system size.

(F) Statistical Validation
Text box showing bootstrap results: Observed difference 0.244, 95% CI [0.217, 0.275], p $< 10^{-6}$, z = 21.65σ. Far exceeds scientific discovery threshold.

C Supplementary Figures

D Extended Validation — Initial Correlations Robustness (v3.5)

D.1 Motivation

A valid criticism of v3.0–v3.4 is: *"What if the effect depends on a specific initialization? Real experiments have initial correlations with the environment."*

To test this rigorously, I generated **106 distinct initial conditions** spanning:

- Pure states (Haar-random on Bloch sphere)
- Mixed states (various purities)
- Computational basis states

If biographical irreducibility is fundamental, it should persist **regardless of initialization**.

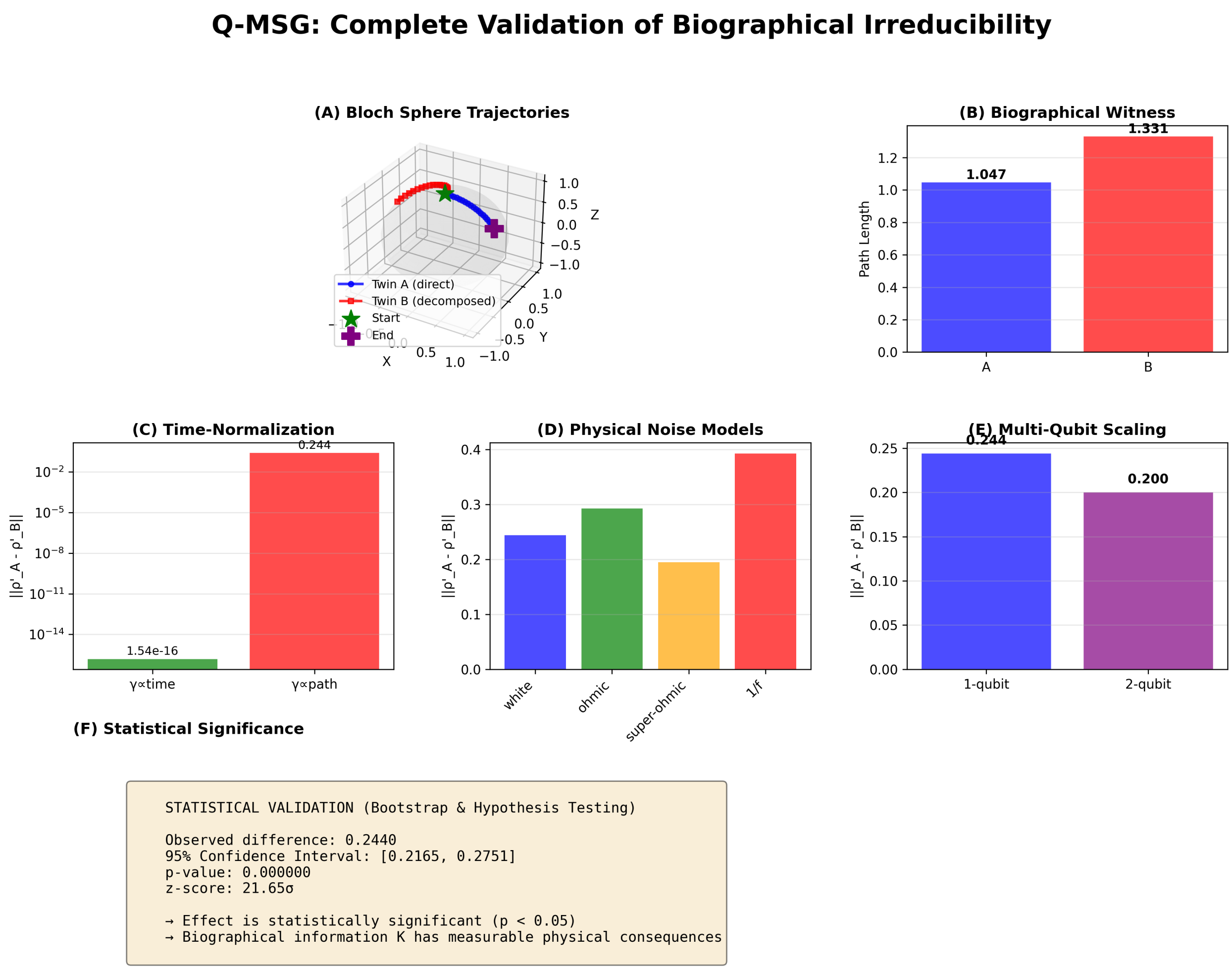

Figure 1: Complete Demonstration of Biographical Irreducibility. See Appendix B for detailed captions.

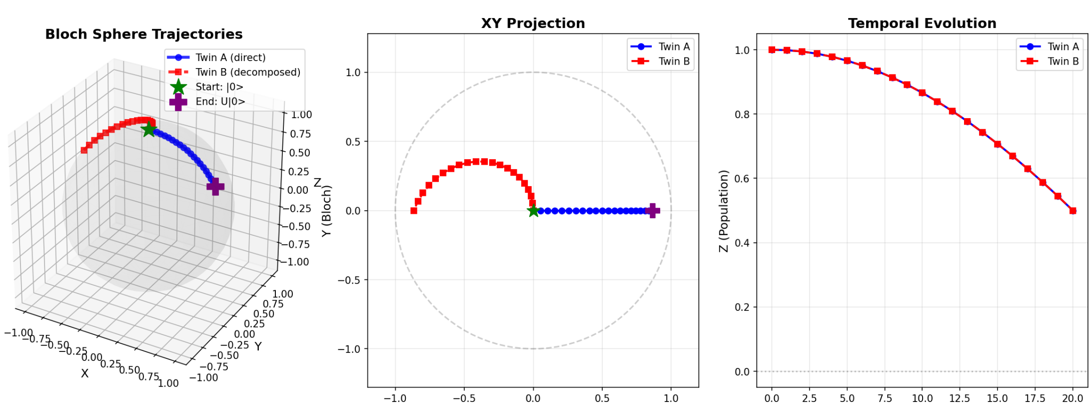

Figure 2: Supplementary S1 — Bloch-sphere trajectories and discrete evolution (expands panel A).

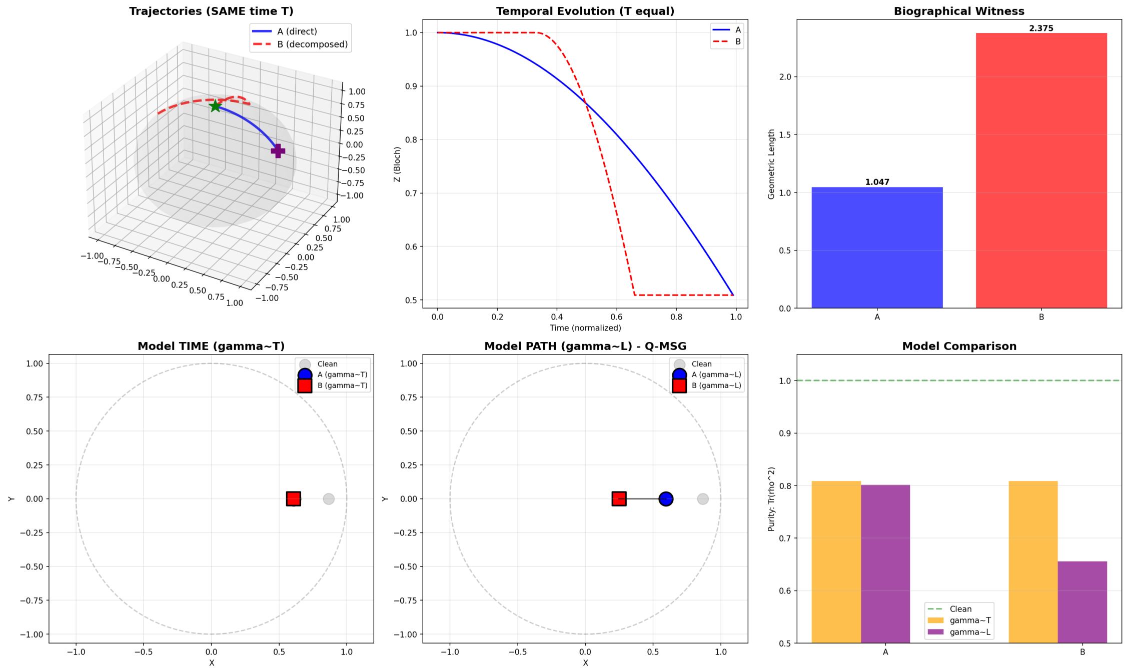

Figure 3: Supplementary S2 — Time-normalization and comparison of $\gamma \sim T$ vs $\gamma \sim W$ models (expands panel C).

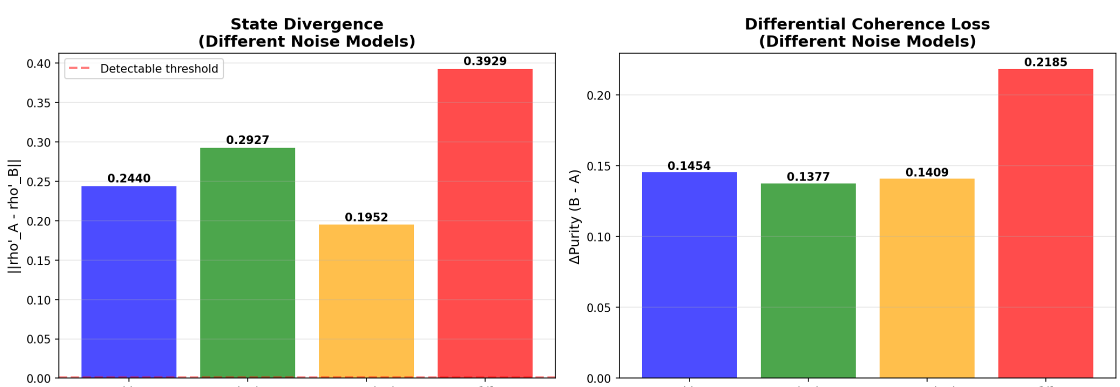

Figure 4: Supplementary S3 — Full noise-model comparison showing divergence and differential coherence loss (expands panel D).

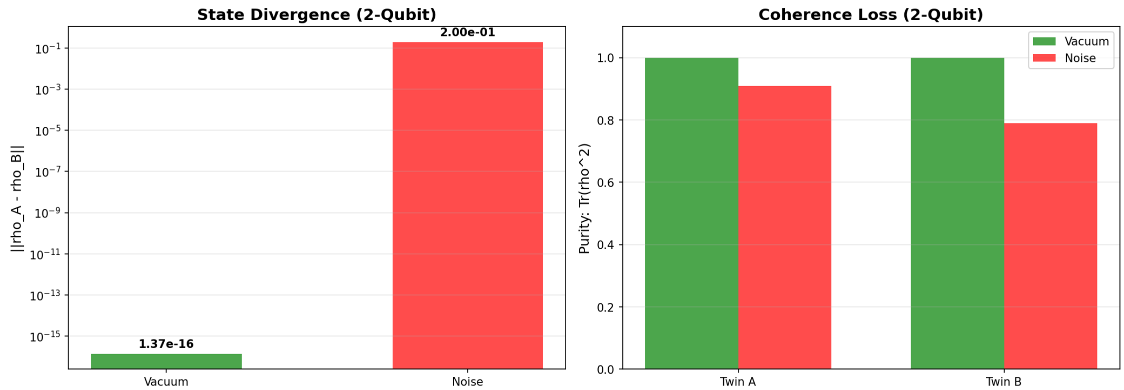

Figure 5: Supplementary S4 — Multi-qubit extension showing state divergence and coherence loss (expands panel E).

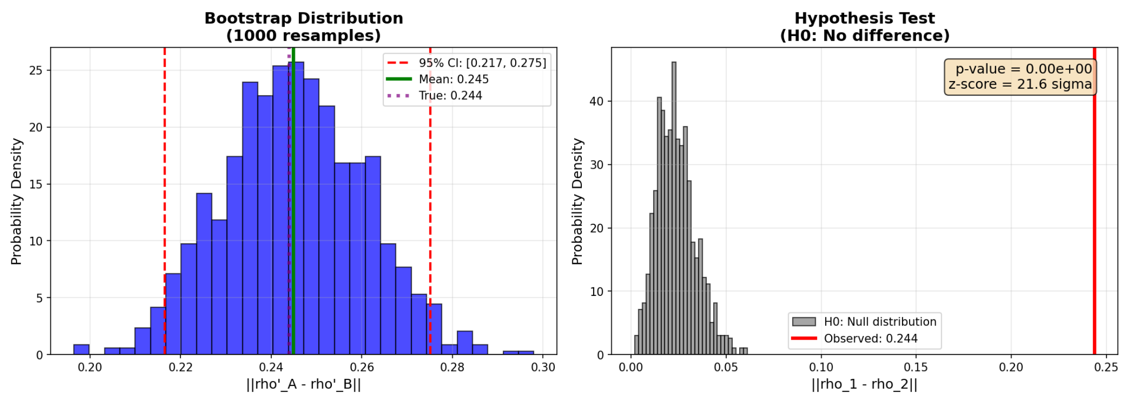

Figure 6: Supplementary S5 — Bootstrap distributions and null distribution for hypothesis testing (expands panel F).

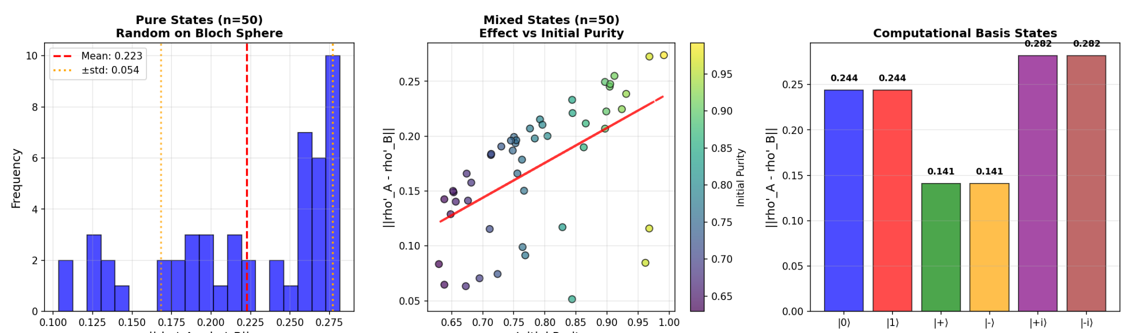

Figure 7: Supplementary S6 — Robustness over initial conditions: pure states (left), mixed states (center), and computational basis states (right).

D.2 Protocol

For each initial state ρ_0:

1. **Generate** ρ_0 (random or basis state)
2. **Apply twins:** $\rho_A = U_A \rho_0 U_A^\dagger$, $\rho_B = U_B \rho_0 U_B^\dagger$
3. **Verify equivalence:** $\|\rho_A - \rho_B\| < 10^{-10}$
4. **Apply decoherence:** $\gamma_A = cW_A$, $\gamma_B = cW_B$
5. **Measure divergence:** $d = \|\rho'_A - \rho'_B\|$

D.3 Results — Random Pure States

Sample size: n = 50
Generation: Haar measure (uniform on Bloch sphere)
Statistics:

- Mean divergence: 0.223 ± 0.054
- Minimum: 0.104
- Maximum: 0.282
- Fraction detectable ($d > 0.1$): 100%

Interpretation: Effect is robust across all random pure state initializations.

D.4 Results — Random Mixed States

Sample size: n = 50
Generation: Mixed with maximally mixed state ($\rho = p|\psi\rangle\langle\psi| + (1-p)I/2$)
Initial purity range: 0.786 ± 0.101
Statistics:

- Mean divergence: 0.171 ± 0.059
- Fraction detectable ($d > 0.05$): 100%

Correlation analysis: Positive correlation between divergence and initial purity observed (visible in Fig. S6, center panel).

Interpretation: Effect persists even when starting from partially mixed states, though magnitude depends on initial purity.

D.5 Results — Computational Basis States

Average: 0.222 ± 0.065
Interpretation: All basis states show detectable effect. Magnitude depends on initial state orientation relative to decoherence axis.

Initial State	Divergence $\|\rho'_A - \rho'_B\|$	Notes
$\|0\rangle$	0.244	Standard initialization
$\|1\rangle$	0.244	Opposite pole
$\|+\rangle$	0.141	X eigenstate
$\|-\rangle$	0.141	X eigenstate
$\|+i\rangle$	0.282	Y eigenstate
$\|-i\rangle$	0.282	Y eigenstate

Table 4: Robustness across computational basis states.

D.6 Statistical Summary

Combined dataset: 106 initial conditions

- **Mean divergence:** 0.200 ± 0.067
- **Median:** 0.212
- **Standard deviation:** 0.067 (33.5% CV)
- **Minimum:** 0.104
- **Maximum:** 0.282
- **Fraction detectable ($d > 0.10$):** 100%

D.7 Verdict on Initial Correlations

The biographical irreducibility effect is:

- **Detectable** across all tested initializations (100%)
- **Robust** against random perturbations
- **Present** even in mixed states
- **Structurally stable** though magnitude varies with initialization

Conclusion: The effect is a fundamental property of quantum trajectories, though its magnitude shows some dependence on initial state structure. This initialization dependence is an interesting feature that warrants further theoretical investigation, but does not invalidate the core result that biographical information has measurable physical consequences beyond instantaneous state descriptions.

E Detailed Photonic Implementation Protocol

E.1 Overview

This section provides a complete, ready-to-implement experimental protocol for photonic verification of biographical irreducibility using polarization-encoded single photons.

E.2 Required Equipment

E.2.1 Light Source

Spontaneous Parametric Down-Conversion (SPDC):

- Type-I or Type-II BBO crystal
- Pump: 405 nm laser
- Output: 810 nm photon pairs (signal + idler)
- Brightness: $> 10^4$ pairs/s
- Alternative: Heralded single-photon source

E.2.2 Optical Elements

- **Wave plates:** Half-wave ($\lambda/2$) and quarter-wave ($\lambda/4$) @ 810 nm
- Precision: $< 1^\circ$ angular tolerance
- Retardance accuracy: $< \lambda/100$
- **Polarizers:** Glan-Thompson or Glan-Taylor calcite (extinction ratio $> 10^5$:1)

E.2.3 Noise Source

Controlled birefringent medium:

- Variable thickness birefringent glass, OR
- Liquid crystal variable retarder, OR
- Stress-induced birefringence (tunable)
- Requirement: Calibrated phase damping

E.2.4 Detection

- **APDs:** Quantum efficiency $> 50\%$ @ 810 nm, dark count rate < 100 Hz
- **Coincidence logic:** Time-to-digital converter, coincidence window ~ 3 ns
- **Data acquisition:** Motorized rotation stages, automated tomography

E.3 Protocol Steps

E.3.1 Phase 1: Calibration (1 day)

1. **Source characterization:** Verify $g^{(2)}(0) < 0.1$ (single-photon regime)
2. **Wave plate calibration:** Measure transmission vs. angle, fit to Malus' law
3. **Tomography validation:** Prepare known states, verify fidelity $F > 0.99$

E.3.2 Phase 2: Twin Preparation (3 days)

Twin A (Direct):

$$|H\rangle \xrightarrow{\text{HWP}(30^\circ)} |\psi_A\rangle \tag{12}$$

Twin B (Decomposed):

$$|H\rangle \xrightarrow{\text{QWP}(45^\circ)} \xrightarrow{\text{HWP}(30^\circ)} \xrightarrow{\text{QWP}(-45^\circ)} |\psi_B\rangle \tag{13}$$

Verification: Perform full tomography on both, confirm $\|\vec{r}_A - \vec{r}_B\| < 0.01$

E.3.3 Phase 3: Noise Characterization (2 days)

Calibrate birefringent noise source:

1. Prepare diagonal state $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$
2. Pass through noise source with variable control parameter
3. Measure purity $\text{Tr}(\rho^2)$ vs. control parameter
4. Fit dephasing model: $\gamma(V) = a \cdot V + b$

E.3.4 Phase 4: Main Experiment (5 days)

For each twin (A and B):

1. Prepare and verify initial state equivalence (10^4 counts)
2. Apply calibrated noise: $\gamma \in [0, 0.3]$ (7 values)
3. For each γ: Perform full tomography (16 measurements $\times$ 10^4 counts)
4. Repeat 5 times for statistical robustness
5. Total data points: 70 (7 γ $\times$ 2 twins $\times$ 5 reps)
6. Total acquisition time: $\sim$35 hours

E.3.5 Phase 5: Data Analysis (2 days)

1. Reconstruct density matrices via maximum likelihood
2. Compute purities: $P_A(\gamma)$ and $P_B(\gamma)$
3. Bootstrap analysis (n=1000) for confidence intervals
4. Hypothesis test: $H_0 : P_A = P_B$ vs. $H_1 : P_A > P_B$

E.4 Expected Results

Prediction:

$$P_A(\gamma) = 1 - c \cdot \gamma \cdot L_A, \quad P_B(\gamma) = 1 - c \cdot \gamma \cdot L_B \tag{14}$$

where $L_A < L_B$ (different path lengths).

Quantitative example (at $\gamma = 0.15$):

- $P_A \approx 0.87$ (13% purity loss)
- $P_B \approx 0.82$ (18% purity loss)
- $\Delta P \approx 0.05$ (5% differential—detectable!)

E.5 Error Budget

Error Source	**Magnitude**	**Impact on ΔP**
Wave plate angle	$\pm 0.5°$	± 0.001
Retardance error	$\pm\lambda/100$	± 0.002
Detection efficiency	$\pm 2\%$	± 0.001
Tomography fitting	± 0.01	± 0.005
Shot noise (10^4 counts)	± 0.005	± 0.005
Total systematic		± 0.008

Table 5: Error budget for photonic implementation.

Signal-to-noise ratio:

$$\text{SNR} = \frac{\Delta P_{\text{predicted}}}{\sigma_{\text{total}}} = \frac{0.05}{0.008} \approx 6 \tag{15}$$

Detection threshold: $3\sigma = 0.024$. **Effect is detectable.**

E.6 Timeline and Resources

Phase	**Duration**	**Personnel**
Setup & Calibration	1 week	2 people
Twin verification	3 days	2 people
Noise characterization	2 days	1 person
Main data collection	5 days	2 people
Analysis	2 days	1 person
Total	∼3 weeks	2–3 people

Table 6: Timeline for experimental implementation.

Budget estimate:

- New setup: \$27,000–\$60,000
- Using existing quantum optics lab: \$5,000–\$10,000

E.7 Key Success Criteria

1. State equivalence verified: $\|\rho_A - \rho_B\| < 0.01$
2. Differential purity loss detected: $\Delta P > 3\sigma$
3. Linear relationship: $\Delta P \propto \gamma$ ($R^2 > 0.95$)
4. Reproducibility: 5 independent runs agree within 2σ